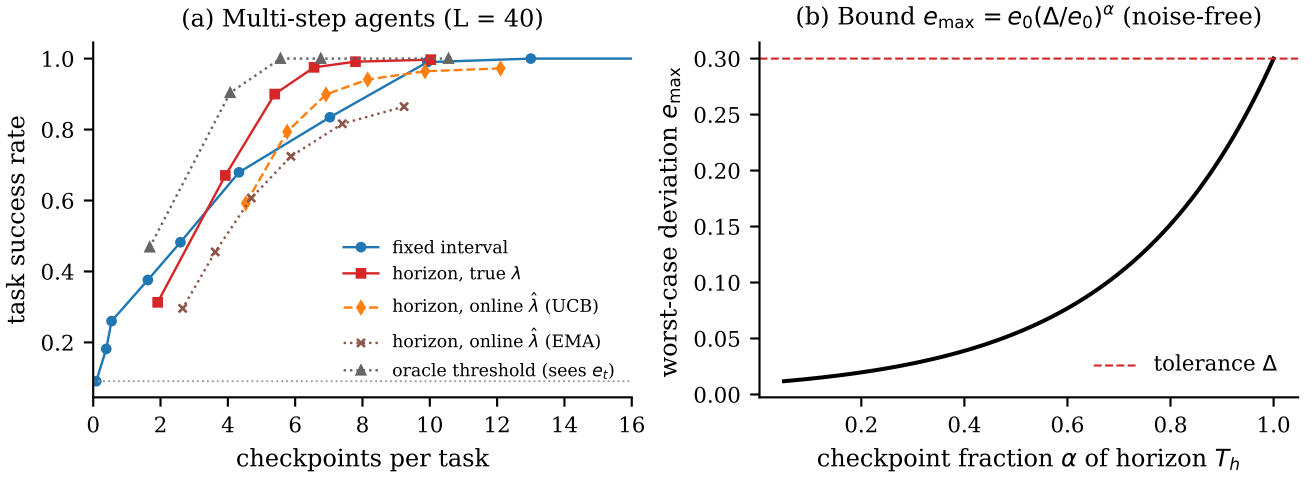
(a) Multi-step agents (L = 40)
task success rate
checkpoints per task
fixed interval
horizon, true λ
horizon, online λ̂ (UCB)
horizon, online λ̂ (EMA)
oracle threshold (sees e_t)
(b) Bound e_max = e_0(Δ/e_0)^α (noise-free)
worst-case deviation e_max
checkpoint fraction α of horizon T_h
tolerance Δ